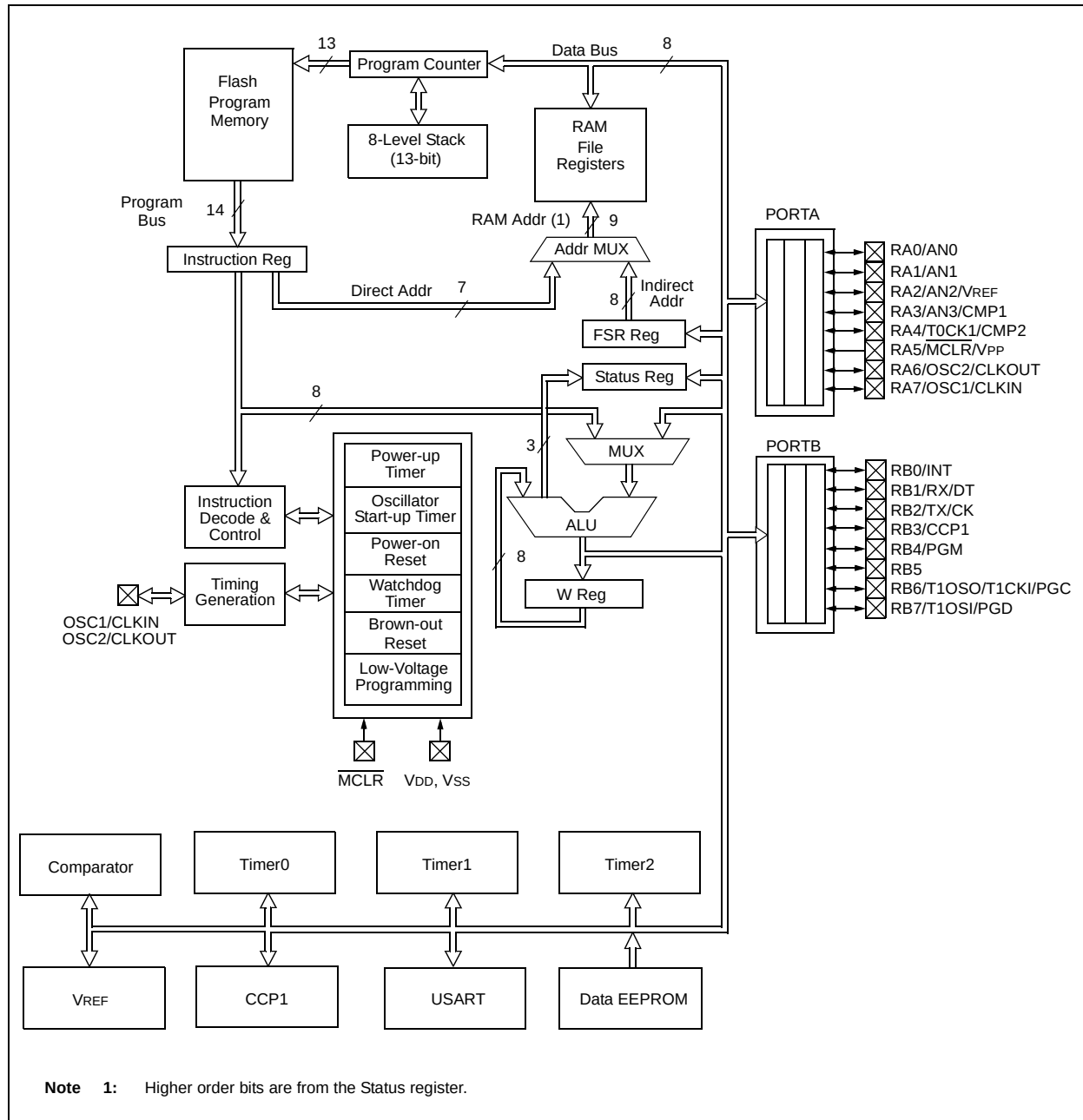


PIC16F627A/628A/648A

FIGURE 3-1: BLOCK DIAGRAM



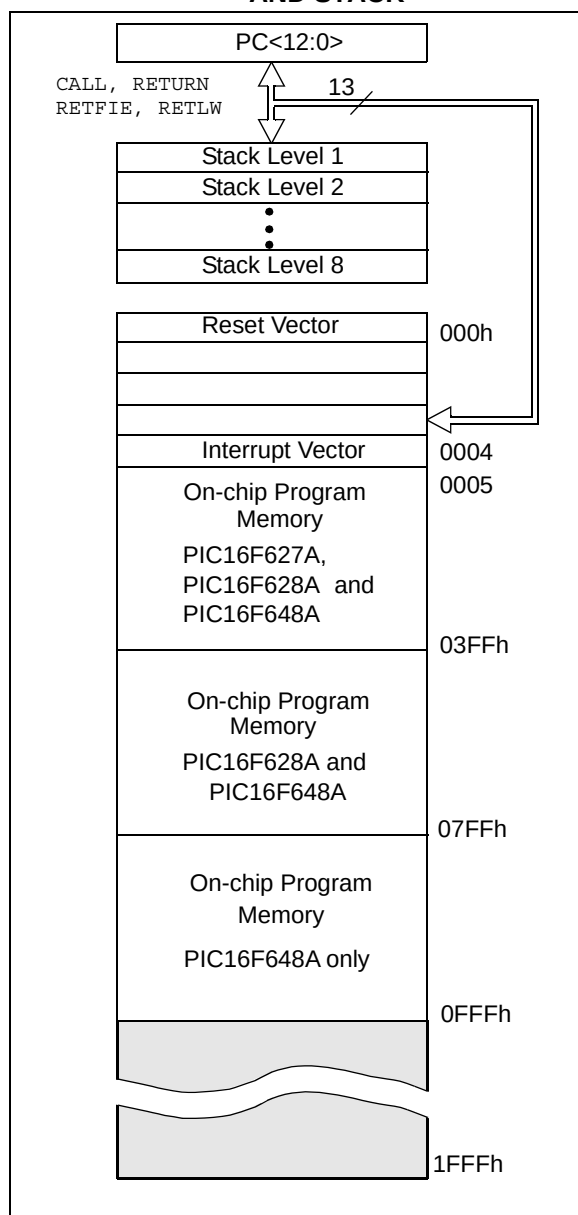
PIC16F627A/628A/648A

4.0 MEMORY ORGANIZATION

4.1 Program Memory Organization

The PIC16F627A/628A/648A has a 13-bit program counter capable of addressing an 8K x 14 program memory space. Only the first 1K x 14 (0000h-03FFh) for the PIC16F627A, 2K x 14 (0000h-07FFh) for the PIC16F628A and 4K x 14 (0000h-0FFFh) for the PIC16F648A are physically implemented. Accessing a location above these boundaries will cause a wrap-around within the first 1K x 14 space (PIC16F627A), 2K x 14 space (PIC16F628A) or 4K x 14 space (PIC16F648A). The Reset vector is at 0000h and the interrupt vector is at 0004h (Figure 4-1).

FIGURE 4-1: PROGRAM MEMORY MAP AND STACK



4.2 Data Memory Organization

The data memory (Figure 4-2 and Figure 4-3) is partitioned into four banks, which contain the General Purpose Registers (GPRs) and the Special Function Registers (SFRs). The SFRs are located in the first 32 locations of each bank. There are General Purpose Registers implemented as static RAM in each bank. Table 4-1 lists the General Purpose Register available in each of the four banks.

TABLE 4-1: GENERAL PURPOSE STATIC RAM REGISTERS

	PIC16F627A/628A	PIC16F648A
Bank0	20-7Fh	20-7Fh
Bank1	A0h-FF	A0h-FF
Bank2	120h-14Fh, 170h-17Fh	120h-17Fh
Bank3	1F0h-1FFh	1F0h-1FFh

Addresses F0h-FFh, 170h-17Fh and 1F0h-1FFh are implemented as common RAM and mapped back to addresses 70h-7Fh.

Table 4-2 lists how to access the four banks of registers via the Status register bits RP1 and RP0.

TABLE 4-2: ACCESS TO BANKS OF REGISTERS

Bank	RP1	RP0
0	0	0
1	0	1
2	1	0
3	1	1

4.2.1 GENERAL PURPOSE REGISTER FILE

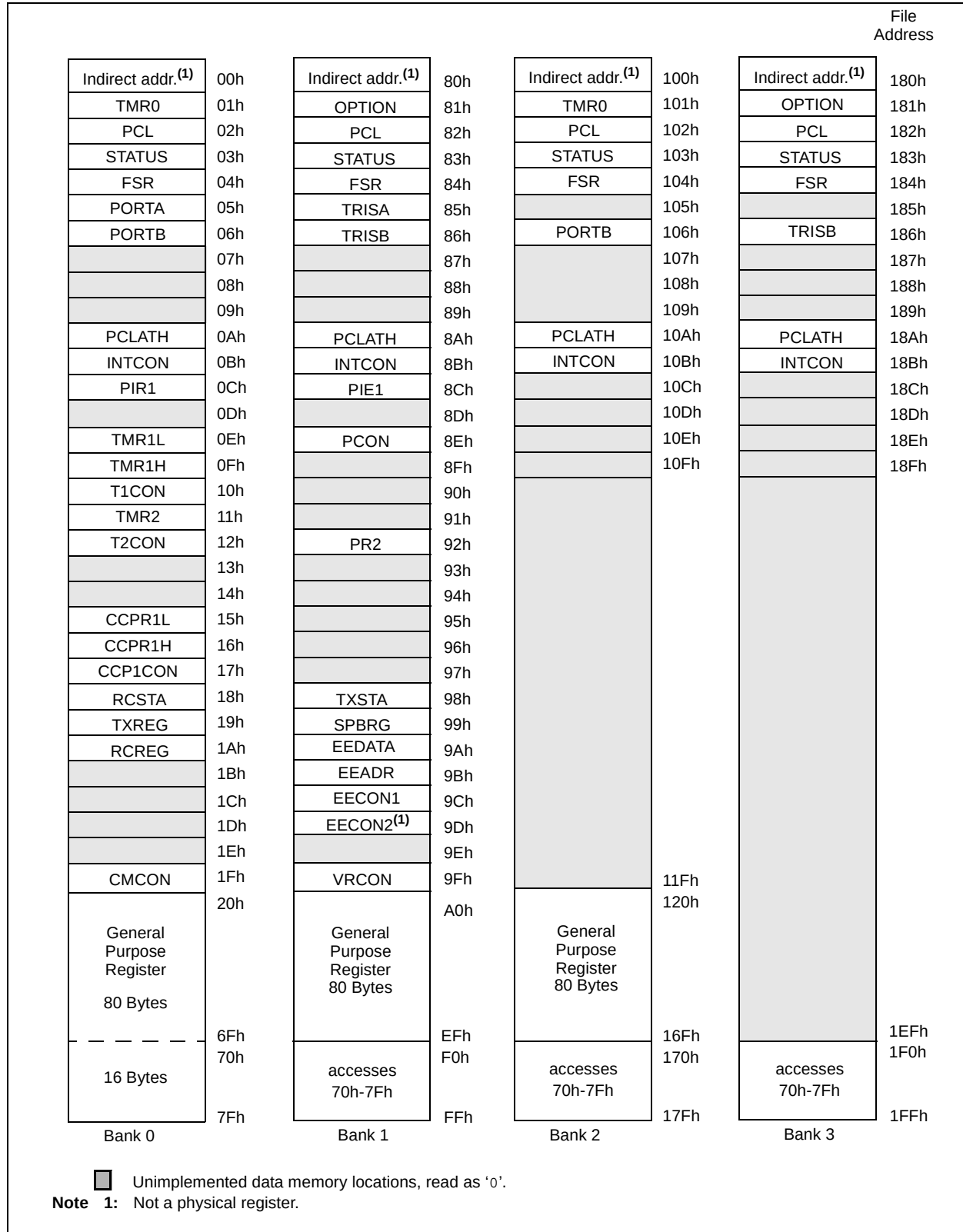
The register file is organized as 224 x 8 in the PIC16F627A/628A and 256 x 8 in the PIC16F648A. Each is accessed either directly or indirectly through the File Select Register (FSR). See **Section 4.4 "Indirect Addressing, INDF and FSR Registers"**.

 Unimplemented data memory locations, read as '0'.

Note 1: Not a physical register.

PIC16F627A/628A/648A

FIGURE 4-3: DATA MEMORY MAP OF THE PIC16F648A



PIC16F627A/628A/648A

4.2.2.1 Status Register

The Status register, shown in Register 4-1, contains the arithmetic status of the ALU; the Reset status and the bank select bits for data memory (SRAM).

The Status register can be the destination for any instruction, like any other register. If the Status register is the destination for an instruction that affects the Z, DC or C bits, then the write to these three bits is disabled. These bits are set or cleared according to the device logic. Furthermore, the $\overline{\text{TO}}$ and PD bits are non-writable. Therefore, the result of an instruction with the Status register as destination may be different than intended.

For example, `CLRF STATUS` will clear the upper-three bits and set the Z bit. This leaves the Status register as "000uu1uu" (where u = unchanged).

It is recommended, therefore, that only `BCF`, `BSF`, `SWAPF` and `MOVWF` instructions are used to alter the Status register because these instructions do not affect any Status bit. For other instructions, not affecting any Status bits, see the "Instruction Set Summary".

Note: The $\overline{\text{C}}$ and DC bits operate as a Borrow and Digit Borrow out bit, respectively, in subtraction. See the `SUBLW` and `SUBWF` instructions for examples.

REGISTER 4-1: STATUS – STATUS REGISTER (ADDRESS: 03h, 83h, 103h, 183h)

R/W-0	R/W-0	R/W-0	R-1	R-1	R/W-x	R/W-x	R/W-x
IRP	RP1	RP0	$\overline{\text{TO}}$	$\overline{\text{PD}}$	Z	DC	C
bit 7							bit 0

- bit 7 **IRP:** Register Bank Select bit (used for indirect addressing)
1 = Bank 2, 3 (100h-1FFh)
0 = Bank 0, 1 (00h-FFh)
- bit 6-5 **RP<1:0>:** Register Bank Select bits (used for direct addressing)
00 = Bank 0 (00h-7Fh)
01 = Bank 1 (80h-FFh)
10 = Bank 2 (100h-17Fh)
11 = Bank 3 (180h-1FFh)
- bit 4 **$\overline{\text{TO}}$:** Time Out bit
1 = After power-up, `CLRWDT` instruction or `SLEEP` instruction
0 = A WDT time out occurred
- bit 3 **$\overline{\text{PD}}$:** Power-down bit
1 = After power-up or by the `CLRWDT` instruction
0 = By execution of the `SLEEP` instruction
- bit 2 **Z:** Zero bit
1 = The result of an arithmetic or logic operation is zero
0 = The result of an arithmetic or logic operation is not zero
- bit 1 **DC:** Digit Carry/Borrow bit (`ADDWF`, `ADDLW`, `SUBLW`, `SUBWF` instructions) (for Borrow the polarity is reversed)
1 = A carry-out from the 4th low order bit of the result occurred
0 = No carry-out from the 4th low order bit of the result
- bit 0 **C:** Carry/Borrow bit (`ADDWF`, `ADDLW`, `SUBLW`, `SUBWF` instructions)
1 = A carry-out from the Most Significant bit of the result occurred
0 = No carry-out from the Most Significant bit of the result occurred
- Note:** For Borrow, the polarity is reversed. A subtraction is executed by adding the two's complement of the second operand. For rotate (`RRF`, `RLF`) instructions, this bit is loaded with either the high or low order bit of the source register.

Legend:

R = Readable bit	W = Writable bit	U = Unimplemented bit, read as '0'
-n = Value at POR	'1' = Bit is set	'0' = Bit is cleared x = Bit is unknown

PIC16F627A/628A/648A

4.2.2.2 OPTION Register

The Option register is a readable and writable register, which contains various control bits to configure the TMR0/WDT prescaler, the external RB0/INT interrupt, TMR0 and the weak pull-ups on PORTB.

Note: To achieve a 1:1 prescaler assignment for TMR0, assign the prescaler to the WDT (PSA = 1). See **Section 6.3.1 “Switching Prescaler Assignment”**.

REGISTER 4-2: OPTION_REG – OPTION REGISTER (ADDRESS: 81h, 181h)

R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
$\overline{\text{RBP}}\text{U}$	INTEDG	T0CS	T0SE	PSA	PS2	PS1	PS0
bit 7							bit 0

- bit 7 **$\overline{\text{RBP}}\text{U}$** : PORTB Pull-up Enable bit
 1 = PORTB pull-ups are disabled
 0 = PORTB pull-ups are enabled by individual port latch values
- bit 6 **INTEDG**: Interrupt Edge Select bit
 1 = Interrupt on rising edge of RB0/INT pin
 0 = Interrupt on falling edge of RB0/INT pin
- bit 5 **T0CS**: TMR0 Clock Source Select bit
 1 = Transition on RA4/T0CKI/CMP2 pin
 0 = Internal instruction cycle clock (CLKOUT)
- bit 4 **T0SE**: TMR0 Source Edge Select bit
 1 = Increment on high-to-low transition on RA4/T0CKI/CMP2 pin
 0 = Increment on low-to-high transition on RA4/T0CKI/CMP2 pin
- bit 3 **PSA**: Prescaler Assignment bit
 1 = Prescaler is assigned to the WDT
 0 = Prescaler is assigned to the Timer0 module
- bit 2-0 **PS<2:0>**: Prescaler Rate Select bits

Bit Value	TMR0 Rate	WDT Rate
000	1 : 2	1 : 1
001	1 : 4	1 : 2
010	1 : 8	1 : 4
011	1 : 16	1 : 8
100	1 : 32	1 : 16
101	1 : 64	1 : 32
110	1 : 128	1 : 64
111	1 : 256	1 : 128

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
 -n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

PIC16F627A/628A/648A

4.2.2.3 INTCON Register

The INTCON register is a readable and writable register, which contains the various enable and flag bits for all interrupt sources except the comparator module. See **Section 4.2.2.4 “PIE1 Register”** and **Section 4.2.2.5 “PIR1 Register”** for a description of the comparator enable and flag bits.

Note: Interrupt flag bits get set when an interrupt condition occurs regardless of the state of its corresponding enable bit or the global enable bit, GIE (INTCON<7>).

REGISTER 4-3: INTCON – INTERRUPT CONTROL REGISTER (ADDRESS: 0Bh, 8Bh, 10Bh, 18Bh)

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-x
GIE	PEIE	TOIE	INTE	RBIE	TOIF	INTF	RBIF
bit 7							bit 0

bit 7	GIE: Global Interrupt Enable bit 1 = Enables all un-masked interrupts 0 = Disables all interrupts
bit 6	PEIE: Peripheral Interrupt Enable bit 1 = Enables all un-masked peripheral interrupts 0 = Disables all peripheral interrupts
bit 5	TOIE: TMR0 Overflow Interrupt Enable bit 1 = Enables the TMR0 interrupt 0 = Disables the TMR0 interrupt
bit 4	INTE: RB0/INT External Interrupt Enable bit 1 = Enables the RB0/INT external interrupt 0 = Disables the RB0/INT external interrupt
bit 3	RBIE: RB Port Change Interrupt Enable bit 1 = Enables the RB port change interrupt 0 = Disables the RB port change interrupt
bit 2	TOIF: TMR0 Overflow Interrupt Flag bit 1 = TMR0 register has overflowed (must be cleared in software) 0 = TMR0 register did not overflow
bit 1	INTF: RB0/INT External Interrupt Flag bit 1 = The RB0/INT external interrupt occurred (must be cleared in software) 0 = The RB0/INT external interrupt did not occur
bit 0	RBIF: RB Port Change Interrupt Flag bit 1 = When at least one of the RB<7:4> pins changes state (must be cleared in software) 0 = None of the RB<7:4> pins have changed state

Legend:

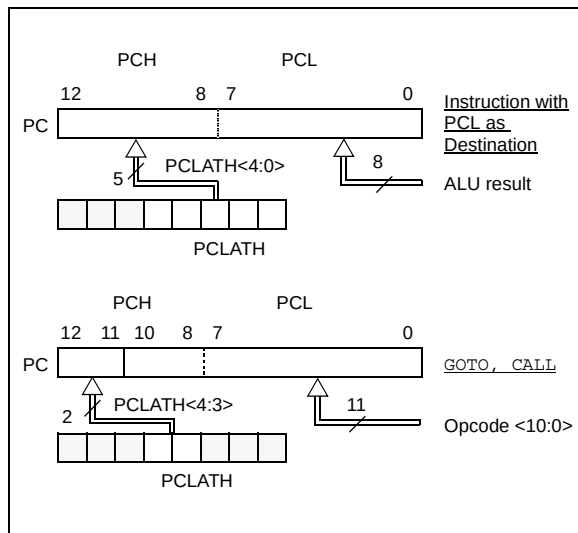
R = Readable bit	W = Writable bit	U = Unimplemented bit, read as '0'
-n = Value at POR	'1' = Bit is set	'0' = Bit is cleared x = Bit is unknown

PIC16F627A/628A/648A

4.3 PCL and PCLATH

The Program Counter (PC) is 13-bits wide. The low byte comes from the PCL register, which is a readable and writable register. The high byte (PC<12:8>) is not directly readable or writable and comes from PCLATH. On any Reset, the PC is cleared. Figure 4-4 shows the two situations for loading the PC. The upper example in Figure 4-4 shows how the PC is loaded on a write to PCL (PCLATH<4:0> → PCH). The lower example in Figure 4-4 shows how the PC is loaded during a CALL or GOTO instruction (PCLATH<4:3> → PCH).

FIGURE 4-4: LOADING OF PC IN DIFFERENT SITUATIONS



4.3.1 COMPUTED GOTO

A computed GOTO is accomplished by adding an offset to the program counter (ADDWF PCL). When doing a table read using a computed GOTO method, care should be exercised if the table location crosses a PCL memory boundary (each 256-byte block). Refer to the Application Note AN556 "Implementing a Table Read" (DS00556).

4.3.2 STACK

The PIC16F627A/628A/648A family has an 8-level deep x 13-bit wide hardware stack (Figure 4-1). The stack space is not part of either program or data space and the Stack Pointer is not readable or writable. The PC is PUSHed onto the stack when a CALL instruction is executed or an interrupt causes a branch. The stack is POPed in the event of a RETURN, RETLW or a RETFIE instruction execution. PCLATH is not affected by a PUSH or POP operation.

The stack operates as a circular buffer. This means that after the stack has been PUSHed eight times, the ninth PUSH overwrites the value that was stored from the first PUSH. The tenth PUSH overwrites the second PUSH (and so on).

Note 1: There are no Status bits to indicate stack overflow or stack underflow conditions.

2: There are no instructions/mnemonics called PUSH or POP. These are actions that occur from the execution of the CALL, RETURN, RETLW and RETFIE instructions, or the vectoring to an interrupt address.

4.4 Indirect Addressing, INDF and FSR Registers

The INDF register is not a physical register. Addressing the INDF register will cause indirect addressing.

Indirect addressing is possible by using the INDF register. Any instruction using the INDF register actually accesses data pointed to by the File Select Register (FSR). Reading INDF itself indirectly will produce 00h. Writing to the INDF register indirectly results in a no-operation (although Status bits may be affected). An effective 9-bit address is obtained by concatenating the 8-bit FSR register and the IRP bit (STATUS<7>), as shown in Figure 4-5.

A simple program to clear RAM location 20h-2Fh using indirect addressing is shown in Example 4-1.

EXAMPLE 4-1: INDIRECT ADDRESSING

	MOVLW	0x20	;initialize pointer
	MOVWF	FSR	;to RAM
NEXT	CLRF	INDF	;clear INDF register
	INCF	FSR	;inc pointer
	BTFSS	FSR,4	;all done?
	GOTO	NEXT	;no clear next
			;yes continue